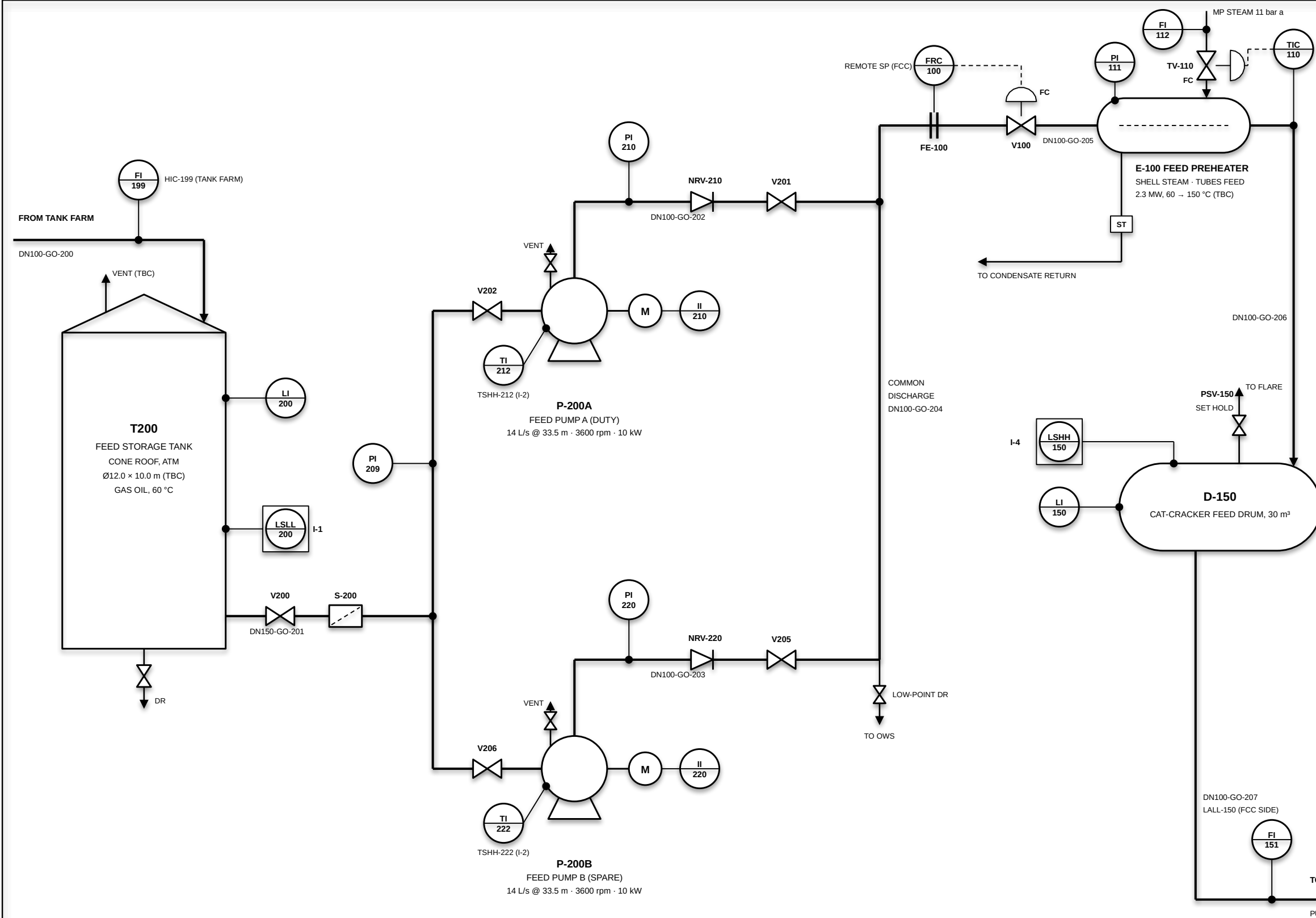




LEGEND

HAND GATE VALVE (V200, V201, V202, V205, V206)

SWING CHECK VALVE, FLOW → (NRV-210/220)

CONTROL VALVE, DIAPHRAGM, FC = FAIL CLOSED

STRAINER

RESTRICTION ORIFICE / FLOW ELEMENT

ELECTRIC MOTOR DRIVER (PUMP: CIRCLE + BASE)

LOCAL INSTRUMENT (NO DCS SIGNAL)

DCS INSTRUMENT (SHARED DISPLAY)

SIS / TRIP FUNCTION

INSTRUMENT SIGNAL

PROCESS LINE, FLOW DIRECTION

UTILITY / DRAIN / VENT LINE

CONNECTION (TEE)

LINE NUMBER: DNxxx-SVC-**nnn**

GO = GAS-OIL FEED · SIZES INDICATIVE (TBC)

MP STEAM / CONDENSATE: UTILITY LINES

PUMP DATA (CASE)

	P-200A	P-200B
SPEED, rpm	3600	3600
MOTOR, kW	10	10
SHUT-OFF HEAD, m	45.0	46.5
RATED	14 L/s	14 L/s
ΔH @ 14 L/s, m	32.0	33.5

GENERAL NOTES
1. DESIGN BASIS: 14 L/s TO THE CAT CRACKER; P-200A/B IDENTICAL, 3600 rpm, 10 kW (UPRATED FROM 1800 rpm).
SHUT-OFF HEAD A 45.0 m, B 46.5 m. B RUNNING: PI-220 34 m, IDLE PI-210 0.5 m, V100 OUTPUT = 80 %.

2. FLUID GAS OIL, 870 kg/m³, 60 °C, Pv 0.3 kPa: ASSUMED FOR THE OTS (TBC). HEADS ON THIS SHEET ARE IN METRES OF LIQUID (1 m = 8.5 kPa).

3. T200 SIZE, SUCTION LOSSES (3.5 m AT 14 L/s), DELIVERY STATIC + BACK PRESSURE (24 m), E-100 AND D-150 DATA: ASSUMED (TBC).

4. PI-209, PI-210, PI-220 ARE LOCAL GAUGES ONLY, PI-210/220 SIT BETWEEN PUMP AND CHECK VALVE: THEY READ THE CASING, NOT THE HEADER.

5. PRESSURE INSTRUMENTS C/W ROOT & TEST VALVES (TYP., NOT SHOWN). PUMP CASING DRAINS TO CLOSED DRAIN (TYP., NOT SHOWN).

6. V100 AND TV-110 FAIL CLOSED. NO BLOCK / BYPASS AROUND V100 ON THE AS-FOUND DRAWING (TBC).

7. FRC-100 SET POINT IS REMOTE FROM THE CAT CRACKER (14 L/s). TIC-110 NORMALLY IN MAN.

8. LINE SIZES INDICATIVE ONLY (TBC). PSV-150 SET PRESSURE AND FLARE TIE-IN: HOLD.

9. DESIGN POINT (A RUNNING): PI-209 0.5 m, PI-210 32.4 m, FRC-100 14 L/s, V100 84 %, II-210 13.2 A, TI-110 150 °C.

INTERLOCKS AND OPERATING NOTES
I-1 LSLL-200 (10 %, 5 s) TRIPS P-200A AND P-200B (LATCHED). LSL-200 (15 %) BLOCKS PUMP STARTS.
I-2 TSHH-212 / TSHH-222 (110 °C, 5 s, ON THE TI-212 / TI-222 ELEMENT) TRIP THEIR OWN MOTOR.
I-3 MOTOR OVERLOAD 49-210 / 49-220 AT THE MCC (21.7 A, 10 s) TRIP THEIR OWN MOTOR.
I-4 LSHH-150 (95 %) FORCES V100 CLOSED (LATCHED UNTIL RESET BELOW 90 %). ESD STOPS BOTH PUMPS, CLOSES V100 AND TV-110.
A. START-UP: SUCTION OPEN, CASING VENTED FULL, DISCHARGE CLOSED; CONFIRM SHUT-OFF HEAD ON THE LOCAL GAUGE; OPEN DISCHARGE WITHIN 2 min.
B. REV A (AS FOUND): SPARE P-200B LINED UP WITH ITS SUCTION V206 AND DISCHARGE V205 OPEN.
C. PROPOSED UPGRADES (REV C, SEE SIS TAB): PIT-210/220 TO DCS WITH PAH ON THE IDLE CASING; MOV DISCHARGE VALVES INTERLOCKED WITH THE MOTORS. NOT SHOWN ON THIS SHEET.
D. THE SIMULATOR SCREEN P&ID IS THE REFERENCE FOR THE DCS, DECKS AND EXERCISES; THIS SHEET IS AN ALTERNATIVE VERSION.

REV	DESCRIPTION	DATE
A	AS FOUND · ISSUED FOR OTS	25-SEP-26
B	ALTERNATIVE VERSION · EPC STANDARD	09-OCT-26

CHEM ENG CHULALONGKORN UNIVERSITY · OTS

P&ID P-200A/B · ALTERNATIVE VERSION

DRAWN: APINAN SOOTITTANTAWAT	CHK'D: --	APP'D: --	DATE: 09-OCT-26
DWG NO. PPS-PID-001	REV B	SCALE N.T.S.	SHEET 1 OF 1